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Review

# Nutrition Rich Fruits and Vegetables: Status and Opportunities in the Era of Genome Editing

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## Abstract

Improvement of nutrition in fruits and vegetables is one of the most important aspects of plant breeding programmes. Biofortification of food crops is necessary to alleviate the challenges of hidden hunger. Consumption of fruits and vegetables provide nutrients like vitamins, minerals, dietary fibres, antioxidants to some extent, however, they either lack or have insufficient amount of several nutrients essential for human health. Recently developed genome editing tools like CRISPR/Cas9 have been successfully utilized for improving the nutrient content of various vegetables and fruits. Considering the potential of the genome editing technique, it is possible to enhance the nutrients as well as to synthesize novel health promoting compounds in the fruits or vegetables by metabolic engineering. This article describes the status and opportunities of improving the nutrition of fruits and vegetables using genome editing approaches.

**Keywords**: Nutrition, CRISPR/Cas9, fruits, vegetables, genome editing

## Introduction

Traditional breeding programmes were mainly focused on improvements in yield which gradually resulted in loss of nutrition from the improved cultivars. Therefore, breeding for nutritionally enriched crops is of prime importance to ensure global nutritional security. Moreover, the mainly consumed cereals and vegetables do not essentially provide all the nutrients required in human diet. As a result, humans are facing the challenge of micronutrient deficiency or hidden hunger. It is also one of the UN sustainable development goals (Goal 2: end the hunger, achieve food security and improved

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nutrition and promote sustainable agriculture) (Lowe, 2021). Therefore, to meet up the challenge of hidden hunger, crops need to be biofortified with micronutrients and other essential nutrients. Several plant foods have antinutritional factors such as saponins, tannins, phytic acid, gossypol, lectins, protease inhibitors, amylase inhibitor, goitrogens, etc. (Samtiya et al., 2020). Reduction of these antinutritional factors in crop species is an important goal of the breeders, however, it is highly difficult to achieve through traditional breeding approaches. The cultivated crops and their crossable species are lacking in nutritional values and therefore, breeders have largely relied on transgenic approaches for enhancing the nutrition e.g. golden rice. However, it has not reached to consumer's plate owing to its tag of genetically modified (GM) crop (Wu et al., 2021). Unlike GM crops, the transgene-free genetically edited crops are allowed for commercial cultivation without rigorous regulation in several countries. This has opened a gate for opportunities to develop nutrition rich genome edited crops and their subsequent commercialization.

Earlier genome editing tools like zinc finger nucleases (ZFNs) and transcription activator like effector nucleases (TALENs) paved the way for their applications in crop improvement but their utilization was restricted due to technical limitations. Nevertheless, since the development of artificial RNA guided engineered nucleases using clustered regularly interspaced short palindromic repeats/CRISPR associated 9 (CRISPR/Cas9), the field of genome editing has bloomed enormously. The artificial CRISPR/Cas9 system was developed from the naturally occurring CRISPR based immunity system in *Streptococcus pyogenes* (Jinek et al., 2012). The CRISPR/Cas9 system is mainly used as a highly efficient tool for site directed mutagenesis, however, its utility has been broadened for several applications like epigenome editing, gene activation, gene repression, base editing, RNA editing, nucleic acid detection, chromosome imaging and so on (Adli, 2018). CRISPR/Cas9 mediated genome editing has been proved to be a miracle technology for crop improvement and it has been successfully utilized for enhancing the quality, stress tolerance, crop yield and also for metabolic engineering in large number of crops (Karkute et al., 2017; Zhu et al., 2020; Gupta and Karkute, 2021).

Rice and wheat are the major staple food crops fulfilling the carbohydrate requirement of humans. On the other hand, fruits and vegetables are the vital components of human diet providing various essential nutrients leading to healthy life. The rich diversity of fruits and vegetables in colours, flavours and tastes also attracts the consumers. Therefore, breeders are aiming at enhancing the nutrition and quality of fruits and vegetables using the most efficient biotechnological tool of genome editing and there are several reports where it has been achieved successfully (Table 1).

## Nutritionally enhanced vegetable crops

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**Table 1:** Status of genome editing mediated improvement of nutrition in vegetable and fruit crops.

| Crop | Target gene | Target phenotype | Reference |
|---|---|---|---|
| **Vegetable crops** | | | |
| Potato | *Vinv* | reduced levels of acrylamide | Clasen et al., 2016 |
| | *16DOX* | decreased steroidal glycoalkaloid content | Nakayasu et al., 2018 |
| | *PPO2* | decreased browning | González et al., 2020 |
| Tomato | *SlANT1* | purple tomatoes | Čermák et al., 2015 |
| | *CRTISO or PSY1* | orange tomatoes and yellow tomatoes, respectively | Dahan-Meir et al., 2018 |
| | *MYB12* | Pink tomatoes | Deng et al., 2018 |
| | *SGR1, LCY-E, Blc, LCY-B1, LCY-B2* | 5.1-fold increase in the lycopene content | Li et al., 2018 |
| | *SlDDB1, SlDET1, SlCYC-B* | increased carotenoid, lycopene, and β-carotene | Hunziker et al., 2020 |
| | *GGPI* | increased vitamin C content | Li et al., 2018 |
| | *cycB* | high lycopene content | Zsögön et al., 2018 |
| | *SlGAD2 and SlGAD3* | sevenfold to 15-fold increase in GABA accumulation | Nonaka et al., 2017 |
| | *SlANT2, SlAN2-like* | decreased anthocyanin content | Yan et al., 2020 |
| | *HYS* | decreased anthocyanin content | Qiu et al., 2019 |
| | *RIN* | decreased volatile organic compounds | Ito et al., 2017 |
| | *FLORAL4* | increased phenylalanine-derived volatile content | Tikunov et al., 2020 |
| | *L1L4* | SSC, fiber, fructose, ascorbic acid, total phenol, carotene, oxalic acid | Gago et al., 2017 |
| Sweet potato | *GBSSI* | decreased amylose content | Wang et al., 2019 |
| | *SBEII* | decreased amylopectin content; increased amylose content | Wang et al., 2019 |
| Brinjal | *PP04, PPOS, and PP06* | decreased browning | Maioli et al., 2020 |
| Mushroom | *PPO* | decreased browning | Waltz, 2016 |
| Lettuce | *LsGGP2* | increased ascorbate content | Zhang et al., 2018 |
| | *SOC1* | Late bolting/flowering | Choi et al., 2022 |
| Chinese kale | *BoaCRTISO* | yellow color of Chinese kale with improved market prospects | Sun et al., 2020 |
| Carrot | *F3H* | decreased anthocyanin content | Klimek-Chodacka et al., 2018 |
| **Fruit crops** | | | |
| Grape | *IdnDH* | decreased tartaric acid content | Ren et al., 2016 |
| Pomegranate | PgUGT84A23 and PgUGT84A24 | unique accumulation of gallic acid 3-0- and 4-0-glucosides | Chang et al., 2019 |
| Banana | *LCYe* | increased F-carotene content | Kaur et al., 2020 |

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Vegetables are an important source of diversified nutrients and health promoting compounds in the human diet. They are rich source of vitamins, minerals, phytochemicals, antioxidants and dietary fibre making them a protective food for several chronic diseases like cancer, obesity, metabolic disorders, etc. (Ulger et al., 2018). This makes vegetable crops the prime target for biofortification to nutritionally charge them with added micronutrients and health promoting compounds through metabolic engineering using genome editing approaches. Globally, the most widely consumed vegetables are potato and tomato; hence their biofortification would lead to diminishing the hidden hunger on larger scale. Production of acrylamide in starchy foods like potato after baking and its subsequent oral intake in humans has a serious health concern (Rommens et al., 2008). Similarly, steroidal glycoalkaloid in potato are also potential toxic compounds. Therefore, reduced levels of acrylamide and steroidal glycoalkaloid is a desirable trait in potato. First genetically edited potato for enhancing the nutritive value was developed in 2016 for reduced level of acrylamide by targeting the *Vinv* gene involved in the accumulation of reducing sugars which cause acrylamide accumulation (Clasen et al., 2016). Subsequently, CRISPR/Cas9 mediated knock-out of *16DOX* gene in potato was achieved to develop the cultivar with decreased content of steroidal glycoalkaloid (Nakayasu et al., 2018). Recently, potato with reduced browning has been developed by mutating the *PPO2* gene that converts phenolic substrates to quinones (González et al., 2020).

Tomato is one of the most attractive vegetables for metabolic engineering and enhancing the nutritional value. Thus, it is the topmost crop that has been subjected to genome editing in 14 different experiments for various traits of nutrition and quality. Majority of the genome edited traits in tomato are for lycopene content or production of different coloured tomatoes. Modification of carotenoid metabolic pathway by targeting different enzymes created tomatoes with different colours. Knock-out of *CRTISO* or *PSY1* genes formed orange tomatoes and yellow tomatoes, respectively (Dahan-Meir et al., 2018) whereas mutation of *MYB12* gene involved in flavonoids metabolic pathway developed pink tomatoes (Deng et al., 2018). Apart from knock-out of the genes, CRISPR/Cas9 has been successfully utilized for upregulation of the *SlANT1* gene involved in anthocyanin biosynthesis by targeting the promoter region (Čermák et al., 2015). This resulted in the enhanced production of anthocyanin imparting purple colour to the tomatoes. On the other hand, tomato fruits with decreased anthocyanin content were developed by knocking out the *SlANT2*, *SlAN2-like* genes (Yan et al., 2020; Zhi et al., 2020) and *HYS* gene (Qiu et al., 2019). Lycopene is an important nutrient in tomato fruits and several attempts have been made through genome editing to enhance lycopene content. Simultaneous knock-out of *SGR1*, *LCY-E*, *Blc*, *LCY-B1*, and *LCY-B2* genes of carotenoid metabolic

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pathway produced tomatoes with 5.1-fold increase in the lycopene content (Li et al., 2018). A similar multiplex editing approach was utilized recently where *SlDDB1, SlDET1, SlCYC-B* genes were subjected to target-AID base editing that produced tomatoes with increased carotenoid, lycopene, and β-carotene (Hunziker et al., 2020).

It is noteworthy to mention the de novo domestication of wild tomato *Solanum pimpinellifolium* using the CRISPR/Cas9 mediated genome editing approach (Zsögön et al., 2018). Here, targeting the *cycB* gene involved in metabolic pathway resulted in almost 500% increase in lycopene accumulation. Genome editing mediated de novo domestication of wild species enables us to develop crops with desirable quality and yield traits while maintaining the wild traits such as biotic and abiotic stress tolerance unaltered. In another attempt to domesticate the *S. pimpinellifolium,* editing the uORF of GGPI gene involved in vitamic C metabolism demonstrated the increased level of foliar vitamin C content (Li et al., 2018). Earlier, knock-out of a single gene *L1L4* in tomato by ZFN developed plants with altered profile of SSC, fiber, fructose, ascorbic acid, total phenol, carotene and oxalic acid (Gago et al., 2017).

Quality of the tomato fruits has been improved by knock-out of *RIN* gene that controls fruit ripening via ethylene (Ito et al., 2017). The resulting mutant plants produced fruits with decreased volatile organic compounds. Tomato is also a fascinating crop for production of new metabolites, amino acids, vitamins, etc. that are crucial for human nutrition. Recently, genome edited tomato fruits with 7 to15 fold increased γ-aminobutyric acid (GABA) content were developed by CRISPR/Cas9 mediated knock-out of *SlGAD2 and SlGAD3* genes (Nonaka et al., 2017). GABA amino acid acts as a neurotransmitter and blocks impulses between nerve cells in the brain. Its oral consumption also lowers the blood pressure and promotes relaxation. The 'Sicilian Rouge' tomato with increased GABA content developed by Sanatech Seed company in Japan is the first genetically edited crop allowed for commercial cultivation (Waltz, 2022). Among the vitamin deficiencies in humans, Vitamin D deficiency is the global health problem which leads to several disorders like cancer and neurocognitive decline. Vitamin D3 is mainly obtained from fish and dairy products while plant-based foods are its poor source. In one of the most exciting applications of CRISPR/Cas9 technology, a genome edited biofortified tomato has been developed that can be used as plant-based source of Vitamin D3 to mitigate the problem of its deficiency (Li et al., 2022).

Apart from major vegetable crops potato and tomato, genome editing has widened its applications for altering the quality and nutrition in several other vegetables like sweet potato, brinjal, carrot, lettuce, Chinese kale, etc. and the list will expand in near future. Being an important source of starch, sweet potato has been modified for improvement of starch quality using genome editing. CRISPR/Cas9 mediated mutation of *GBSS1* gene functioning in granule-bound starch

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biosynthesis produced tubers with decreased amylose content whereas mutation of *SBEII* gene involved in starch branching for amylopectin developed tubers with increased amylose content but decreased amylopectin (Wang et al., 2019). Browning of mushrooms due to oxidation of phenolic compounds by polyphenol oxidase (PPO) enzyme is an undesirable trait. Reduced activity of PPO could result in decreased browning. The non-browning or decreased browning mushrooms were successfully developed by CRISPR/Cas9 mediated mutation of *PPO* gene that became the first genome edited product to be commercialized for human consumption (Waltz et al., 2016). Similarly, knock-out of *PPO4*, *PPO5* and *PPO6* genes that convert phenolic substrates to quinones using CRISPR/Cas9 technique has been carried out in brinjal to develop fruits with decreased browning quality (Maioli et al., 2020).

Leafy vegetables like lettuce are potential crops for metabolic engineering due to their high edible biomass. Their nutrition content and quality have been improved through genome editing. Recently, CRISPR/Cas9 mediated single base mutation of *SOC1* gene in lettuce has been carried out to develop late bolting plants (Choi et al., 2022). Ascorbic acid content of lettuce has been increased to 150% by editing the uORF of *LsGGP2* gene involved in vitamin C biosynthesis (Zhang et al., 2018). In addition to enhanced vitamin C content, these edited plants showed oxidation stress tolerance as well. In case of Chinese kale, editing the *BoaCRTISO* gene involved in carotenoid biosynthesis produced yellow colour of Chinese kale with improved market prospects (Sun et al., 2020).

## Nutritionally enhanced fruit crops

Genome editing in fruit crops for improving the nutrition and quality is not yet widely exploited, although it has been used for improving yield and stress tolerance. Fruit crops are the main source of human nutrition including vitamins, minerals, dietary fibres and several health promoting bioactive compounds. Pomegranate is a rich source of phenolic metabolites and therefore, it can be exploited for production of novel and beneficial phenolic compounds through metabolic engineering. It has been demonstrated recently by targeting UDP-dependent glycosyltransferases (UGTs) enzyme encoding genes *PgUGT84A23* and *PgUGT84A24* with overlapping activities in ß-glucogallin biosynthesis (Chang et al., 2019). This resulted in unique accumulation of gallic acid 3-0- and 4-0-glucosides. This paved the way for synthesis of novel compounds in fruits by multiplex editing. Banana is one of the most widely consumed fruits and thus, its biofortification would benefit large population. Kaur et al., (2020) developed banana with increased β--carotene content by knocking out the *LCYe* gene involved in the β--carotene metabolism pathway. CRISPR/Cas9 mediated genome editing has also been successfully applied in grapes where targeted mutagenesis of *IdnDH* gene resulted in reduced tartaric acid content (Ren et al., 2016). Apart from improving nutrition and quality, the genome editing has been demonstrated in several fruit crops like apple,

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sweet orange, fig, watermelon, strawberry, kiwifruit, groundcherry, etc. (Kaur et al., 2020).

## Conclusion

After the development of CRISPR/Cas9 tool, genome editing has become the most efficient and widely used technique for crop improvement. Apart from yield and stress tolerance, improving the nutrition in vegetable and fruit crops is one of the most potential areas where genome editing is being exploited. Fruits and vegetables being the rich source of diverse nutrients can be effectively targeted for metabolic engineering to synthesize novel compounds. Number of countries including India has decided not to regulate the transgene-free genome edited crops developed through Site-Directed Nuclease-1 (SDN-1) and Site-Directed Nuclease-2 (SDN-2), which has boosted the genome editing based crop improvement programmes. The only limitation of using the genome editing is the non-availability of transformation and regeneration protocol in several crops and therefore, to utilize the potential of this technique, it is of utmost importance to develop such protocols in important fruits and vegetables.

## Conflict of interest

Authors declare no conflict of interest.

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